Hypothesis Testing:

Hypothesis testing is a type of inferential statistics that is used to test assumptions and draw conclusions about the population from the available sample data.

It involves setting up a **null hypothesis** and an **alternative hypothesis**
followed by conducting a statistical test of significance.
A conclusion is drawn based on the value of the

- test statistic,
- the critical value, and
- the confidence intervals.

Null Value Testing/Null Hypothesis:

This is the default or initial assumption. It often represents a statement of no effect, no difference, or no change. It is denoted as H0.

Null hypothesis testing is a formal approach to decide whether a statistical relationship in a sample reflects a real relationship in the population or is just due to chance.

Alternative hypothesis:

This is the statement you want to test or the claim you're investigating. It represents the opposite of the null hypothesis and is what you hope to find evidence for. It is denoted as Ha or sometimes as H1.

The alternative hypothesis is a statement used in statistical inference experiments. **It is contradictory to the null hypothesis.**

Points to be noted for Hypothesis Testing:

If the output value **is within** the Confidence Interval then we can say the **Null hypothesis is True and Accepted**. In this scenario, we can say we fail to reject the Alternate Hypothesis.

If the output value **is not within** the Confidence Interval then we can say **Null HyPothis is Rejcted** and **Alternate Hypothesis gets Accepted.**

Points to be noted for Hypothesis Testing:

If the output value **is within** the Confidence Interval then we can say the **Null hypothesis is True and Accepted.** In this scenario, we can say we fail to reject the Alternate Hypothesis.

If the output value **is not within** the Confidence Interval then we can say **Null HyPothis is Rejcted** and **Alternate Hypothesis gets Accepted.**

Term	What it tells you	Typical value
Confidence level	How confident we want to be in our interval procedure	95%
Significance level (α)	How much evidence we require to reject the null hypothesis(100-CL)	5%
Confidence interval	A range of plausible values for a population parameter	42 to 48

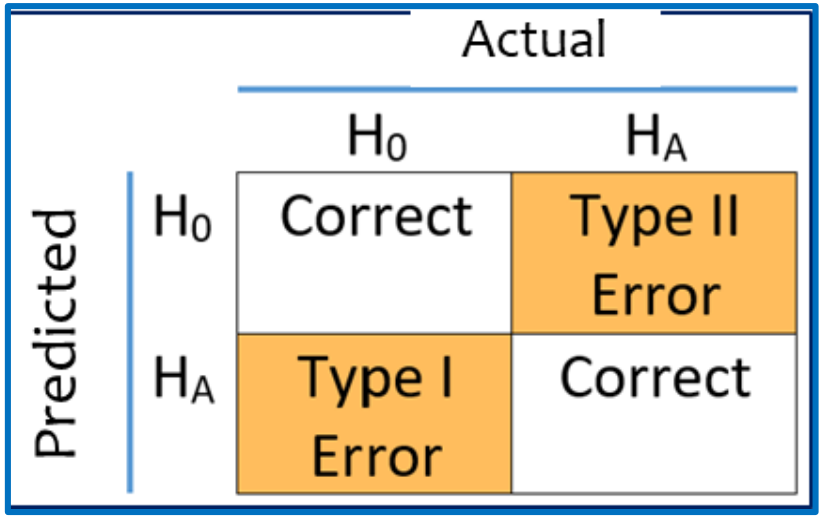
Errors in Hypothesis Testing

Type 1 error :

It happens when the null hypothesis of an experiment should have been Accepted however you accept Alternate Hypothesis is called **Type 1 error or** false positive.

Type 2 error:

It happens where you should have accepted the Alternate Hypothesis but you accept Null Hypothesis.



Which error is considered to be more dangerous?

Type 1 error is considered to be worse or more dangerous than type 2.

Because to reject what is true is dangerous or more harmful than keeping the data that is not true.

		Actual	
		Ho	H1
Predicted	Ho	Ho/Ho	h1/ho
	h1	ho/h1	h1/h1

		Actual	
		Ho	H1
Predicted	Ho	Correct(ok)	Error (Type2)
	h1	Error (Type1)	Correct(ok)

Which error is considered to be more dangerous?

Type 1 error is considered to be worse or more dangerous than type 2.

Because to reject **what is true** is dangerous or more harmful than keeping the data that is not true.

Errors in Hypothesis Testing

Type 1 error :

It happens when the null hypothesis of an experiment should have been Accepted however you accept Alternate Hypothesis is called **Type 1 error or** false positive.

Type 2 error:

It happens where you should have accepted the Alternate Hypothesis but you accept Null Hypothesis.

Step 1: State the Hypotheses

Step 2: Assume the Null Hypothesis Is True

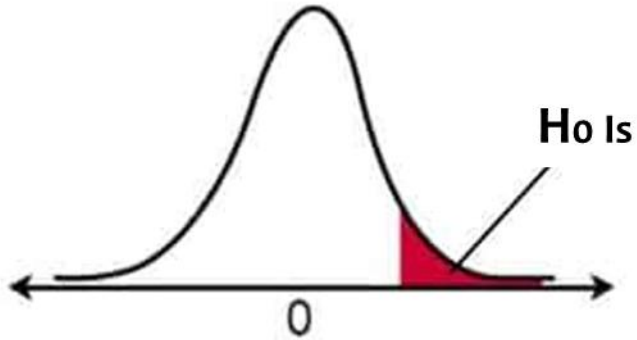
Step 3: Calculate a Test Statistic

Step 4: Use graph to know the rejection status

Plot on Graph to know the H0,h1 status

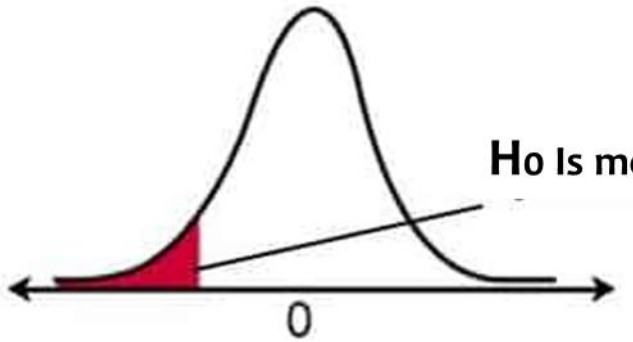
Step 5: Use p-value to know the rejection status

Compare With Significance Level



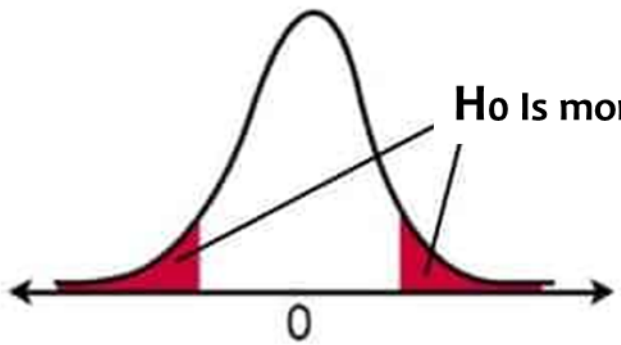
Right-tail test

$H_0: \mu > \text{value}$



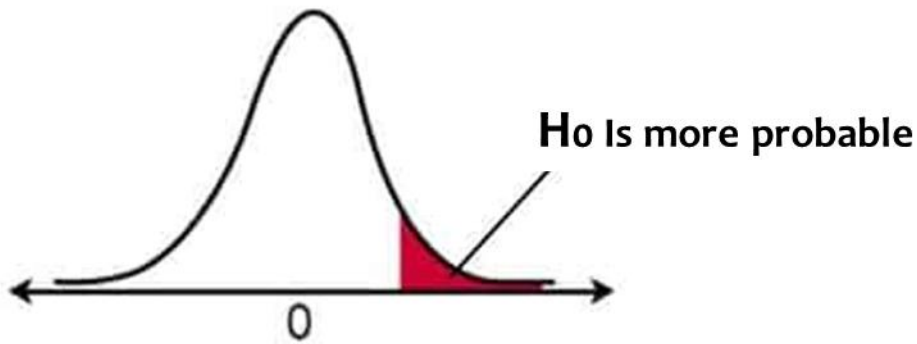
Left-tail test

$H_0: \mu < \text{value}$



Two-tail test

$H_0: \mu \neq \text{value}$



Right-tail test

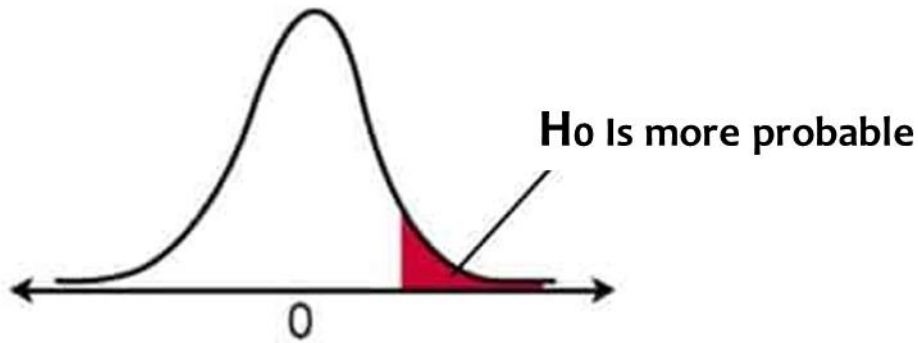
$H_0: \mu > \text{value}$

Problem

A company wants to find out if a new Machine has improved employee productivity.
Before the new Machine, the average productivity was **50 units per day,** and the population standard deviation was **5 units**.
After the use of New Machine, a random sample of **30 employees** had an average productivity of **53 units per day**. The company wants to test whether the New machine has significantly increased productivity.

Population mean:	$\mu = 50$
Sample mean:	$x = 53$
Population standard deviation:	$\sigma = 5$
Sample size:	$n = 30$
Confidence Interval ci	ci= 0.95
Significance Value	α= 0.05

If Final output is inside the rejection area we can say that null hypothesis is rejected.
If the Final output is outside the the rejection or inside the ci area we can say that null hypothesis is accepted.



Right-tail test

$H_0: \mu > \text{value}$

This is a **one-sample right-tailed Z-test** because:
Test Result Value > Claimed Value

Define the Null Hypothesis

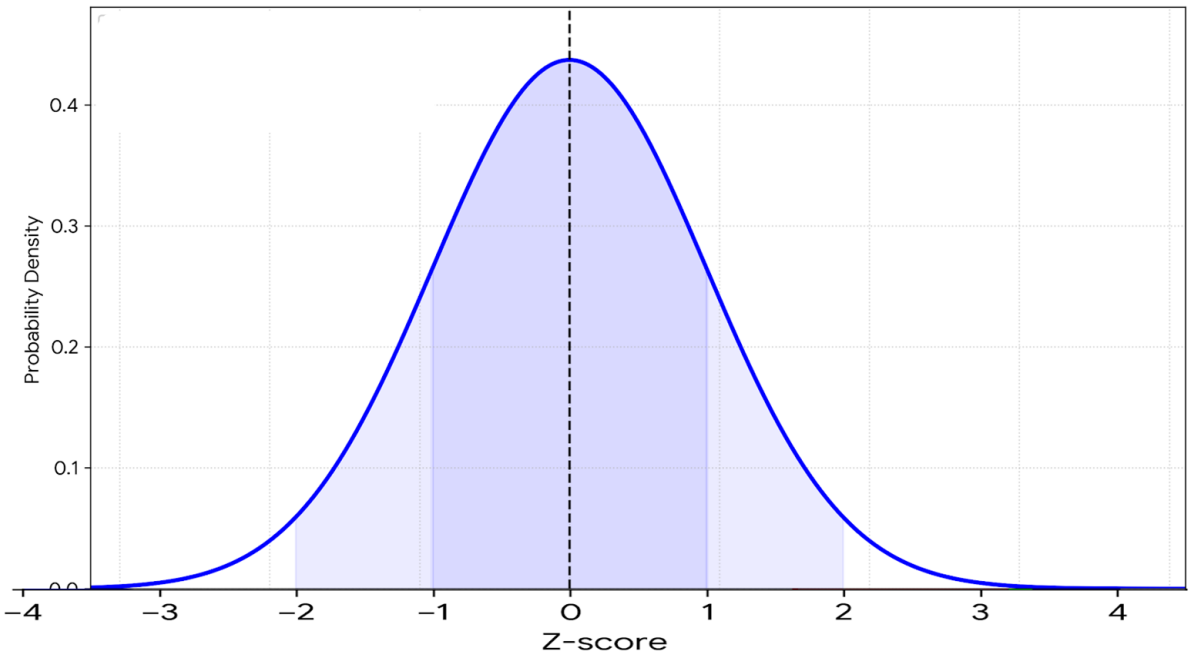
Null Hypothesis (H_0): The New Machine program did not increase productivity.

Ho : μ=50

Alternative Hypothesis (H_1): New Machine program increased productivity.

H1 : Sample Mean>Population mean

H1 : 53>50



Calculate the

1) **Critical Value for Confidence Level(0.95) Significance level(.05) & :**
   to know the rejection Area on graph
2) **Z-score :**
   to know if the value fall under the reaction area or not
1) **Standard Error :**
   Helps to calculate the Z-score.

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$

$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

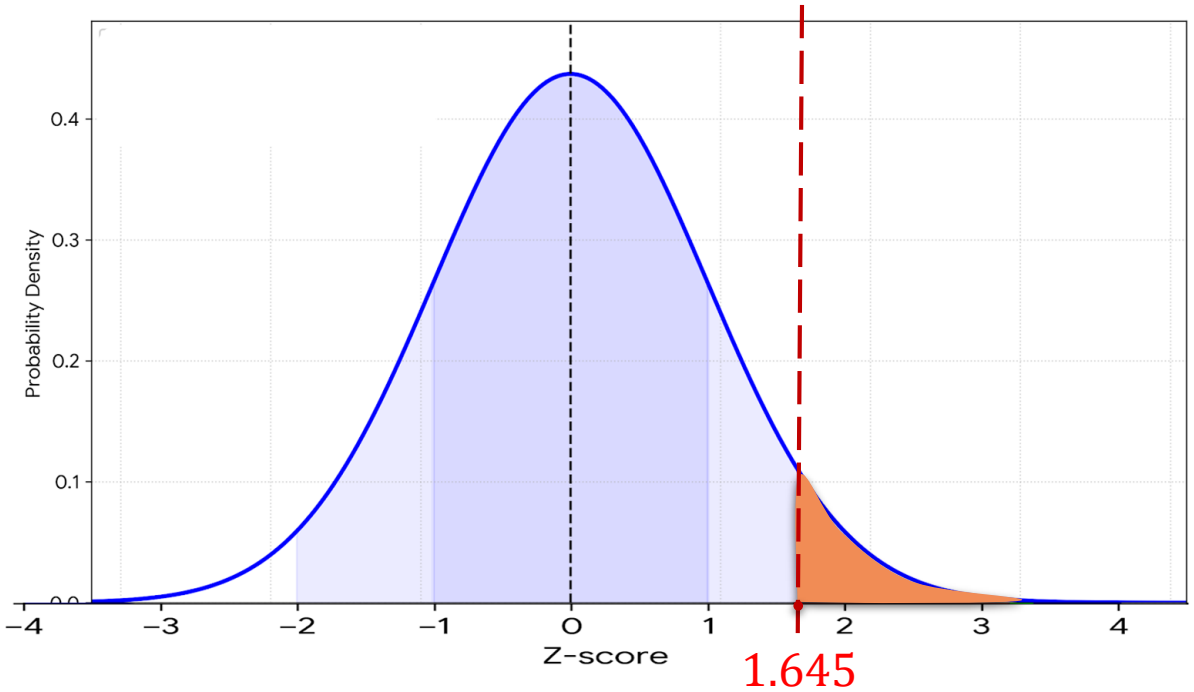
Calculate the Critical Value for Confidence Level(0.95) Significance level(.05)

$$\frac{0.04 + 0.05}{2} = 0.045$$

$$Crtical\ Value = Zvalue + \frac{Value1 + Value2}{2}$$

$$Crtical\ Value = 1.6 + \frac{0.04 + 0.05}{2}$$

$$Crtical\ Value = 1.6 + 0.045 = 1.645$$



Know the Critical Value for 0.95

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169

Z-table (Standard Normal Distribution Table).

Calculate the standard error

Population standard deviation: $\sigma = 5$

Sample size: $n = 30$

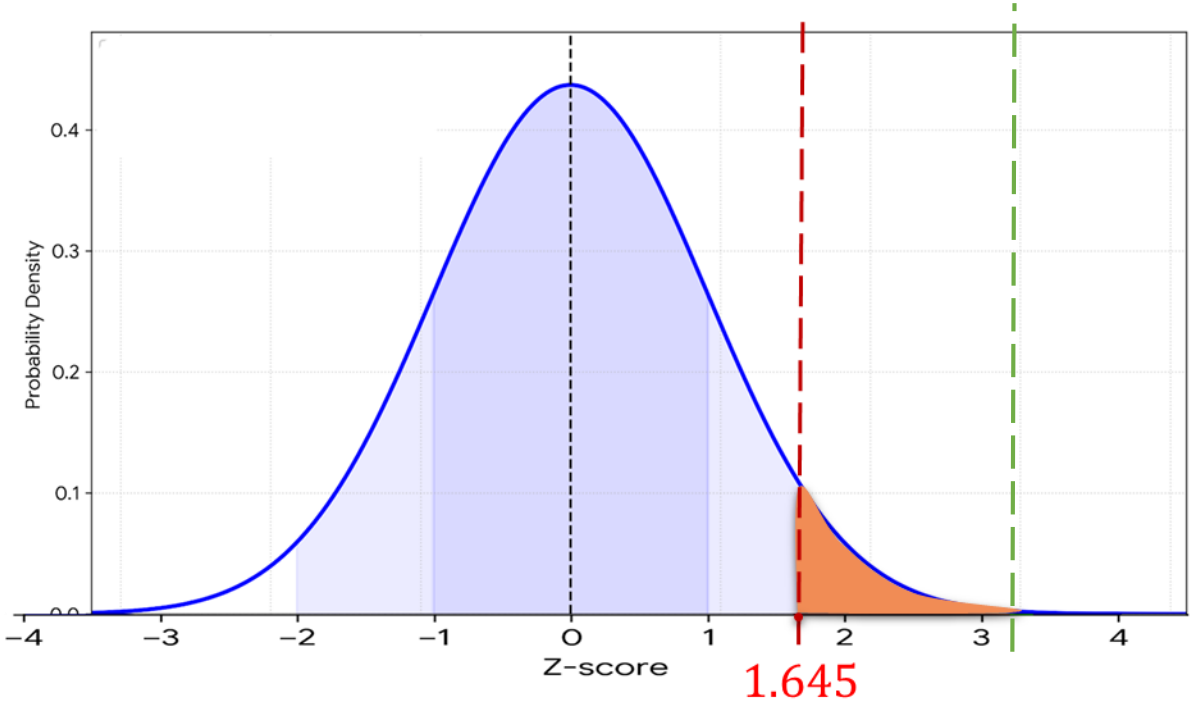
$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

Standard Error: $\frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{30}} \approx 0.913$

Calculate the Z-Score

```
Sample mean:            x    = 53
Population mean:        μ    = 50
Standard Error          (SE)= 0.913
```

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$



Test Statistic (Z): $Z = \frac{53-50}{0.913} = \frac{3}{0.913} \approx \mathbf{3.29}$

Decision: Reject H0 because 3.29 > 1.645.

Conclusion: The New machine increased productivity at the alpha = 0.05 level.

Prove the same using P-Value

$IF\ pvalue <$ significance level

We can say that Null hypothesis is rejected

$\boldsymbol{IF\ pvalue >}$ **significance level**

We can say that Null hypothesis is Accepted

$$pvalue = 1 - "Critical\ Value\ of\ zScore"$$

Calcuate the $the\ Critical\ Value\ of\ zScore$

$$zScore = 3.29$$
$$pvalue = 1 - 0.99950$$
$$pvalue = 0.0005$$

$pvalue(0.0005) <$ significance level(0.05)

We can say that Null hypothesis is rejected

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
+2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
+2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
+2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
+2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
+2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
+2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
+2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
+2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
+2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
+3	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
+3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
+3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
+3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
+3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
+3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
+3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
+3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
+3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
+3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997
+4	.99997	.99997	.99997	.99997	.99997	.99997	.99998	.99998	.99998	.99998

Calculate the critical value for ci or significance level

Plot the critical value on graph to know the rejection Area.

Calculate the SE(Standard Error)

Calculate the z-SCORE

How to know if the Null hypothesis is rejected or accepted.

1) By using Graph
   a. Plot the Z-score on the graph to know if the value fals under the rejction area or acceptance area.
2) By using Numbers
   a. Using Zscore > Significance level
      i. Zscore > Significance level Reject the Null Hypothesis
      ii. Zscore < Significance level Accept the Null Hypothesis
   b. using p-value

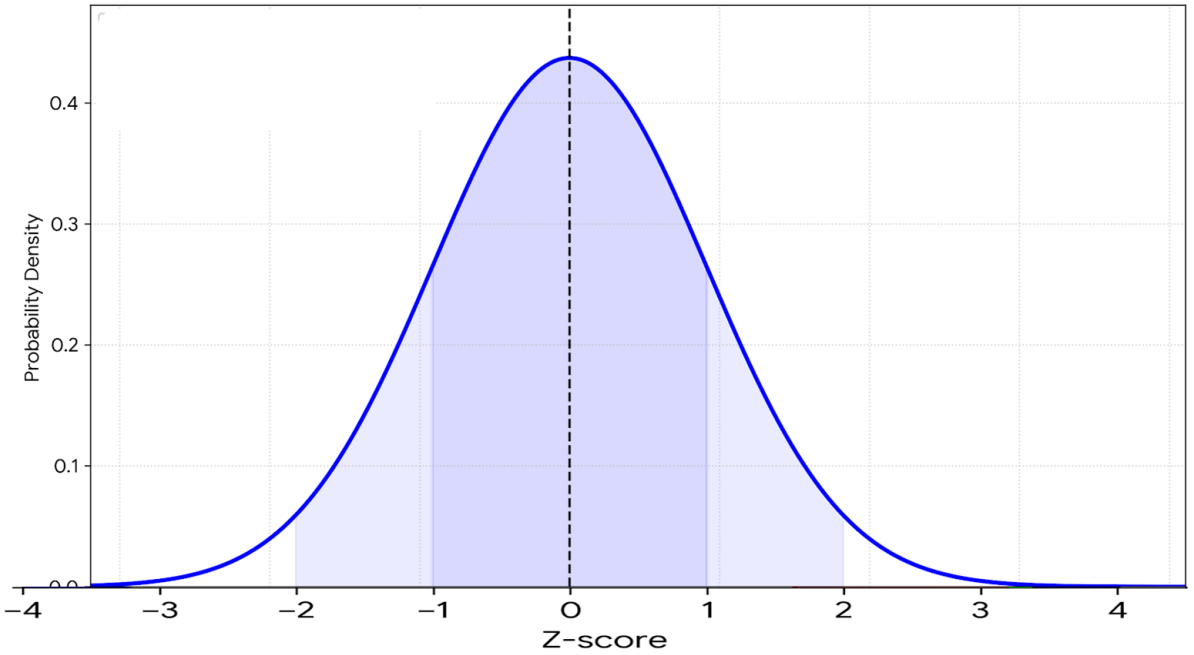
A company wants to find out if a new Machine has improved employee productivity. Before the new Machine, the average productivity was **50 units per day**, and the After the training, a random sample of **30 employees** had an average productivity of **53 units per day**. The company wants to test whether the New machine has significantly increased productivity.

Population mean:	$\mu = 50$
Sample size:	$n = 30$
Sample mean:	$x = 53$
Sample standard deviation:	$\sigma = 4.5$
CI	$= 0.97$
Significance Value	$= 0.03$

Define the Null Hypothesis
This is a **one-sample right-tailed Z-test** because:

Null Hypothesis (H_0): The New Machine program did not increase productivity.
Ho : $\mu=50$

Alternative Hypothesis (H_1): New Machine program increased productivity.
H1 : $\mu>50$



Calculate the

1) **Critical Value for Confidence level(.03) & :**
   to know the rejection Area on graph
2) **Z-score :**
   to know if the value fall under the reaction area or not
1) **Standard Error :**
   Helps to calculate the Z-score.

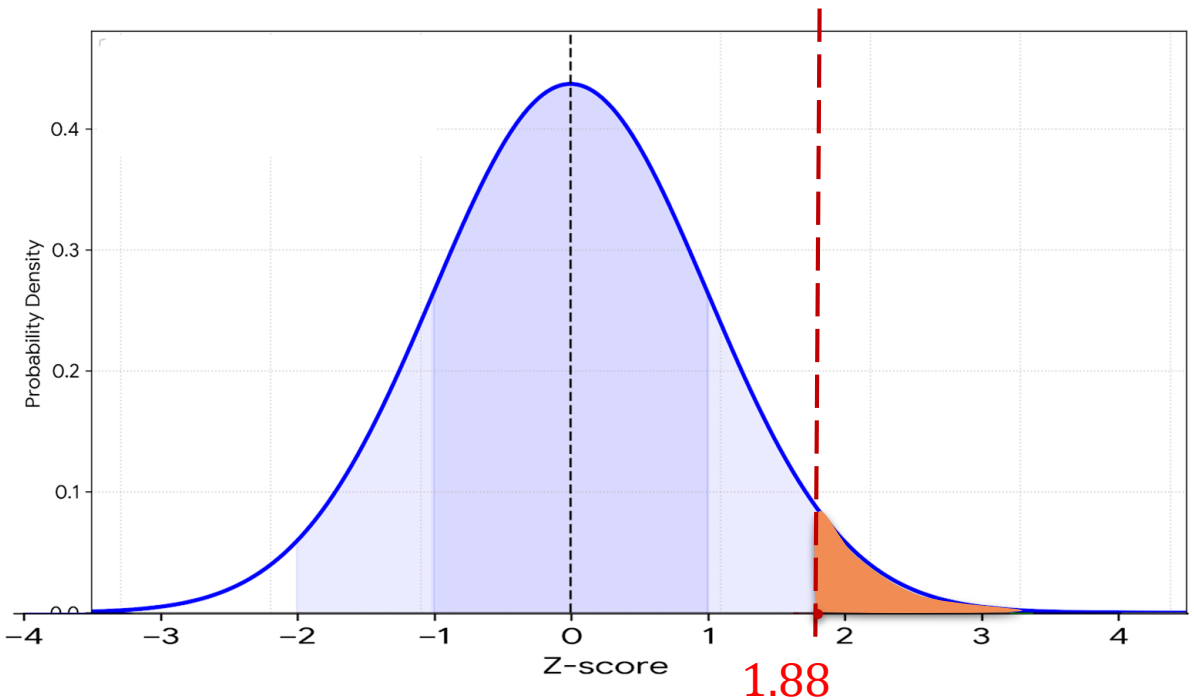
$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$

$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

$$Crtical\ Value = \frac{Value1 + Value2}{2}$$

$$Crtical\ Value = \frac{0.08 + 0.08}{2} = 0.085$$

$$Crtical\ Value = 1.8 + 0.085 = 1.88$$



Know the Critical Value for 0.97

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169

Calculate the standard error

```
Standard deviation: σ = 4.55
Sample size:        n = 30
```

$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

$$standard\ Error\ (SE) = \frac{4.5}{\sqrt{30}}$$

$$standard\ Error\ (SE) = \frac{4.5}{5.5772}$$

$$standard\ Error\ (SE) = 0.8215$$

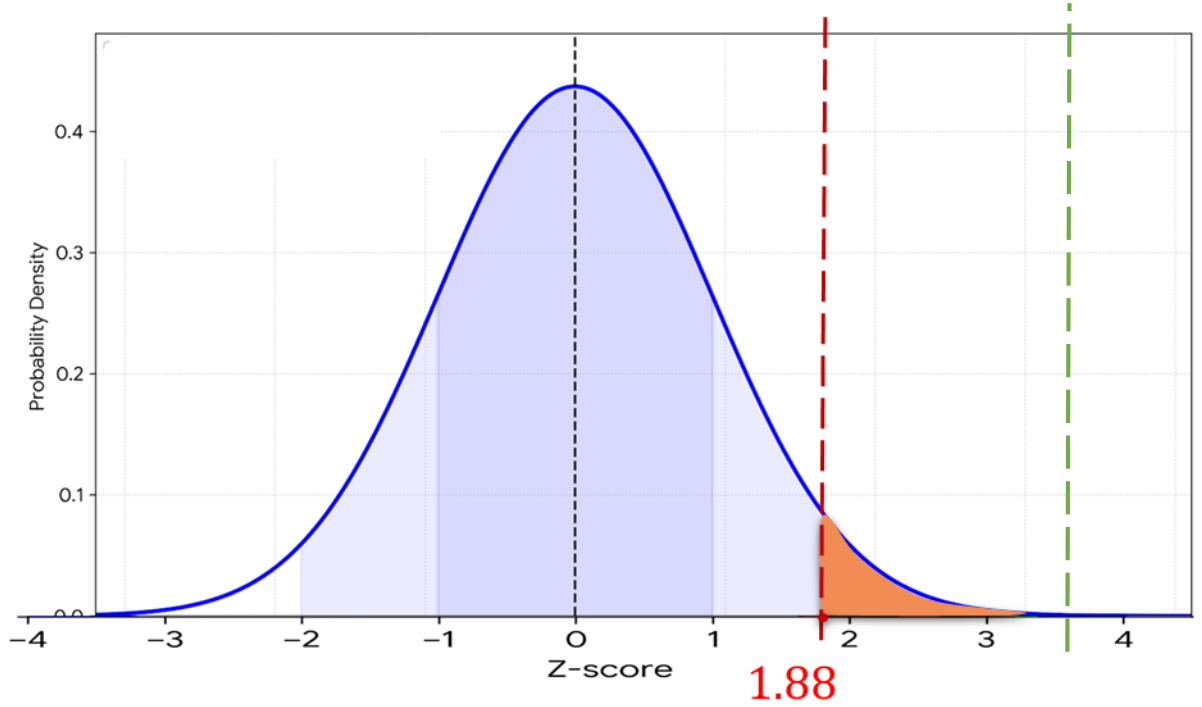
Calculate the Z-Score

```
Population mean:        μ   = 50
Standard Error          SE  = 0.8215
Sample mean:            x   = 53
```

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE} = \frac{53 - 50}{0.8216} = \frac{3}{0.8216}$$

$$\text{Zscore } (Z) = 3.65$$



Decision: Reject Ho because 3.65 > 1.88.

Conclusion: The New machine increased productivity at the alpha = 0.05 level.

$$\text{Zscore}\ (Z) = 3.65$$

$$pvalue = 1 - "Critical\ Value\ of\ zScore"$$

$$pvalue = 1 - 99987$$

$$pvalue = 0.00013$$

$pvalue(0.00013) <$ significance level(0.05)

We can say that Null hypothesis is rejected

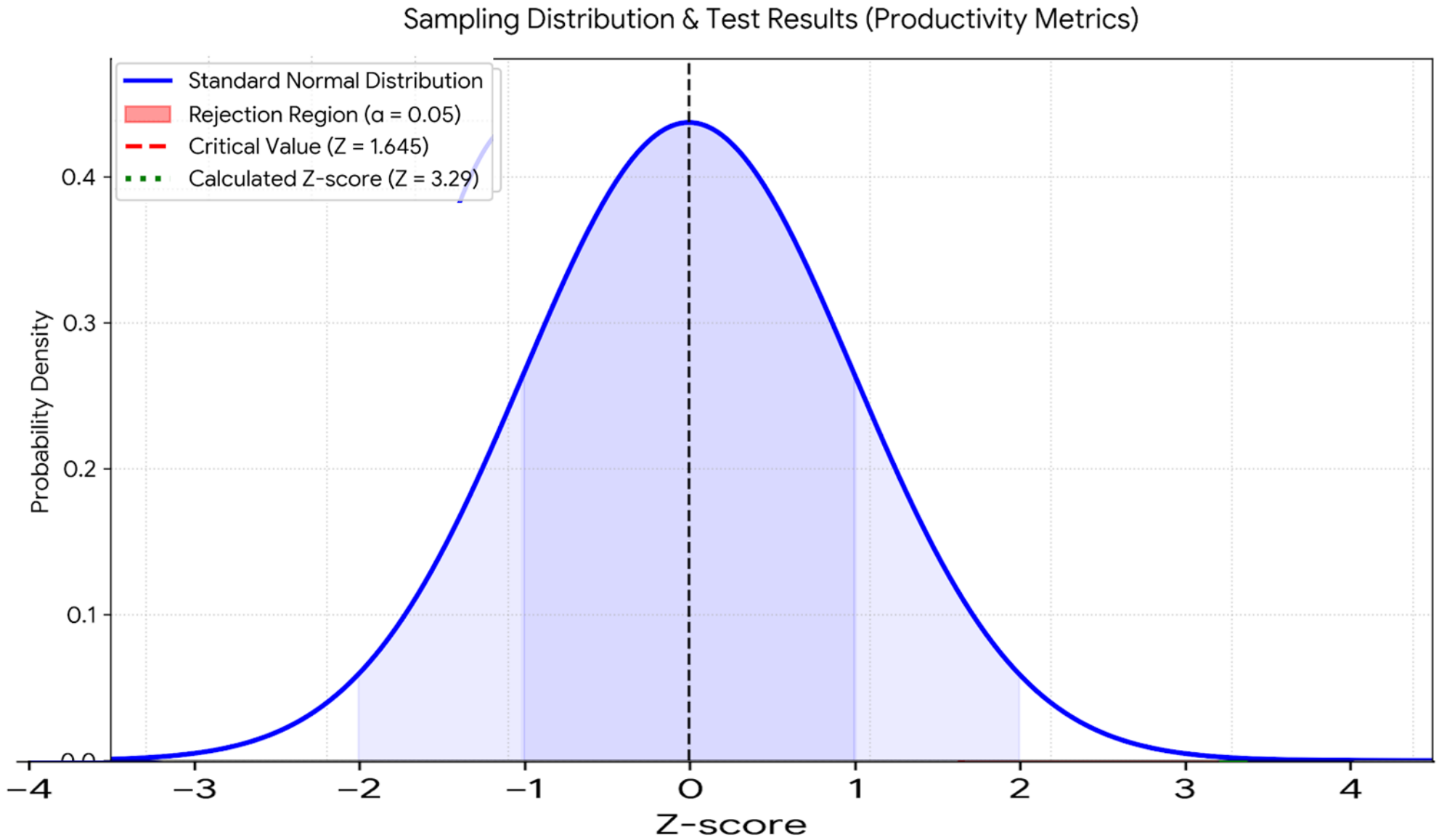
Conclusion

Because the p-value (0.00013) is far lower than your significance level (0.05), the result is highly statistically significant.

There is a mere 0.013 chance that this productivity increase happened.

Null hypothesis is rejected.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
+2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
+2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
+2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
+2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
+2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
+2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
+2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
+2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
+2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
+3	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
+3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
+3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
+3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
+3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
+3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
+3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
+3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
+3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
+3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997
+4	.99997	.99997	.99997	.99997	.99997	.99997	.99998	.99998	.99998	.99998


Sampling Distribution & Test Results (Productivity Metrics)
Standard Normal Distribution
Rejection Region (α = 0.05)
Critical Value (Z = 1.645)
Calculated Z-score (Z = 3.29)
Probability Density
0.4
0.3
0.2
0.1
0.0
−4
−3
−2
−1
0
1
2
3
4
Z-score

Steps in Hypothesis Testing

Step 1: State the Hypotheses

Null Hypothesis (H_0)

The default assumption.

$$H_0 : \mu = 10$$

Meaning:

- The average battery life is **10 hours**.

Alternative Hypothesis (H_1)

The claim being investigated.

$$H_1 : \mu \neq 10$$

Meaning:

- The average battery life is **not 10 hours**.

Step 2: Assume the Null Hypothesis is True

Start by assuming the company's claim is correct.

Ask:

> If the true average battery life is really 10 hours, how likely is it to observe a sample mean as extreme as 9.2 hours?

Worked Example

Given:

- Claimed mean = 10
- Sample mean = 9.2
- Standard Error = 0.25

Calculate z-score

$$z = \frac{9.2 - 10}{0.25}$$

$$z = -3.2$$

A z-score of **−3.2** is far from 0.

Given

$$z = \frac{9.2 - 10}{0.25} = -3.2$$

The alternative hypothesis is:

$$H_1 : \mu \neq 10$$

Since the alternative uses ≠, this is a **two-tailed test**.

Most Z-tables give probabilities for **positive z-values**. First look up:

$$z = 3.20$$

The table gives:

$$P(Z \leq 3.20) = 0.999313$$

Using the symmetry of the normal distribution:

$$P(Z \leq -3.20) = 1 - 0.999313$$

$$= 0.000687$$

$$p = 2(0.000687) = 0.001374 \approx 0.0014$$

Find p-value

$$p \approx 0.0014$$

Decision

$$0.0014 < 0.05$$

Therefore:

✅ **Reject H_0**

Conclusion

There is **strong evidence** that the average battery life is **not 10 hours**.

Errors in Hypothesis Testing

Type 1 error :

it happens when the experiment of **null hypothesis of is true** but rejected often called a false positive.

type 2 error:

it occurs when the experiment **null hypothesis is false** but still not rejected, also known as a false negative.

Which error is considered to be more dangerous?

Type 1 error is considered to be worse or more dangerous than type 2 because to reject what is true is more harmful than keeping the data that is not true.

Hypothesis Testing

Hypothesis testing is a method used to decide whether sample data provides enough evidence to support or reject a claim about a population.

Think of it as a formal way of answering:
"Is this observed result just random chance, or is something real happening?"

Example: Battery Life Claim

Suppose a company claims:
"Our batteries last 10 hours on average."
You test 100 batteries and find:

- Sample mean = 9.2 hours

Now you want to know:
Is the difference (9.2 vs 10) large enough to conclude the claim is false?
This is a hypothesis testing problem.

Null Hypothesis (H0H_0H0)
The default assumption.
H0:μ=10H_0: \mu = 10H0:μ=10
Meaning:
The average battery life is 10 hours.
Alternative Hypothesis (H1H_1H1)
The claim you're investigating.
H1:μ≠10H_1: \mu \ne 10H1:μ□=10

Meaning:
The average battery life is not 10 hours.

Different hypothesis tests are used in Hypothesis Testing.

- Z Test:
- T-Test:
- F Test:
- Annona Test:
- Chi-Square Test:

$$ci = \bar{x} \mp z\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$ci = 70 \mp 1.96\left(\frac{10}{\sqrt{100}}\right)$$

$$ci = 70 \mp 1.96$$

$$ci = 68.04, 71.96$$